



Android Application Development

I have been developing on Android for the past one year, and thought it would be nice to start a series on basic Android application development for beginners. I chose to write for LINUX For You because of its huge reach and popularity with FOSS fans. Here's a basic Android application—a Simple Interest Calculator, which illustrates the application development method. To follow this tutorial, you should have written 'core' Java code, and should have some experience using the Eclipse IDE.

The following steps are a shortened version of what's on the Android developer website; you can see <http://developer.android.com/sdk/installing.html> for more details.

- **Prepare your development computer:** You may need to install JDK version 5 or 6, and Eclipse (version 3.4 or 3.5, but not 3.6) if you don't already have these installed.
- **Download and install the SDK starter package from** <http://developer.android.com/sdk/index.html> and extract its contents to a folder, then add the folder location to your PATH variable (especially if using Windows).
- **Install the ADT (Android Development Tools) plug-in in Eclipse:** Via Help -> Install New Software -> Add, set up the remote update site <https://dl-ssl.google.com/android/eclipse/>. If you have problems with this URL, replace https with http. Install the ADT plug-in (see Figure 1), restart Eclipse, and set the 'Android' preferences in Eclipse to point to the location where you've extracted the SDK.
- **Add Android platforms and other components to your SDK:** Use the Android SDK and AVD (Android Virtual Device) Manager included in the SDK starter

package, which is represented by the Eclipse toolbar button shown in Figure 2. With it, you can add one or more Android platforms that you intend to develop (e.g., Android 2.0 or Android 2.2) and other components to your SDK. You can also install the documentation for the version of Android you are downloading. To launch the Android SDK and AVD Manager, you can also execute the Android tool in the SDK's /tools folder.

Creating a new Android project

Once you've set up the environment, you can create a sample Android Project in Eclipse; the new Android project dialogue is shown in Figure 3. Some important fields in this dialogue:

- **Build Target** lets you choose one of the Android SDK versions you have installed with the SDK and AVD Manager.
- **Application Name** is the title that is shown when you run your application.
- **Package name** must be at least two levels deep (for example, "home.saket").
- The **Activity** name specifies the name of your class (that is going to be created), which will extend the Activity class from the Android library. We will cover the Activity class later in the tutorial.